

# FATIMA MANSOOR

## AI/ML ENGINEER · FULL-STACK ENGINEER

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### SUMMARY

Software Engineer specialising in **Agentic AI architectures** and **full-stack web development**. Experienced in building end to end intelligent applications from LLM powered RAG pipelines and fine tuned ML models to scalable React/Node.js production systems with hands on deployment workflows using Docker, AWS, Vercel, and Streamlit Cloud.

### EDUCATION

#### BE Software Engineering

2022 — Present (Final Semester)

Muhammad Ali Jinnah University, Karachi, Pakistan

#### FSc Pre-Engineering

2020 — 2022

Karachi, Pakistan

### SKILLS

**AI / ML:** TensorFlow · PyTorch · Scikit-learn · LangChain · LangGraph · Hugging Face · OpenCV · LLM Fine-tuning · RAG · Pandas, NumPy · Groq API

**Frontend:** React.js · Next.js · TypeScript · JavaScript · Tailwind CSS · React Native

**Backend:** Node.js · Express.js · FastAPI · REST APIs · Python

**Data / Databases:** MongoDB · SQL · Firebase · Cloudinary

**Infrastructure & Tools:** Docker · AWS · Git · Vercel · Netlify · Linux · Jupyter Notebook · Postman · Figma

### EXPERIENCE

#### Deep Learning / Machine Learning Intern

Nov 2025 — Dec 2025

NextGen, Karachi, Pakistan

- Fine-tuned ML models for document classification; improved model precision through iterative hyperparameter tuning configurations.
- Architected automated data preprocessing pipelines, significantly reducing manual engineering infrastructure overhead.

### FEATURED PROJECTS

#### AI-Powered Job Fit Checker | Agentic AI, Full-Stack

Demo · GitHub

Production system using LangGraph agentic architecture for deep semantic résumé-JD analysis, skill-gap detection, and actionable feedback. Integrated Next.js UI frontend with a responsive FastAPI backend engine.

**Technologies Used:** Next.js, TypeScript, Tailwind CSS, FastAPI, LangGraph, Groq Cloud API

#### Pro Document QA — RAG System | RAG, NLP

Demo · GitHub

Modular RAG application enabling automated document text extraction maps, local HuggingFace embedding indexing, and Groq-hosted LLM execution loops for source-cited conversational query responses.

**Technologies Used:** Streamlit, LangChain, Hugging Face, Groq LLM, Python

#### Customer Churn Prediction | ML, Classification

GitHub

End-to-end predictive analysis workflow containing explicit feature parsing engineering, execution layer evaluation metrics, and comparative model verification routines for tracking churn vulnerabilities.

**Technologies Used:** Python, Scikit-learn, Pandas, NumPy

#### Talent Match — Job Portal | Full-Stack

Demo · GitHub

Full-stack dynamic recruitment engine operating secure token authentication methods, cloud assets storage mapping via Cloudinary, and high-frequency real-time database endpoints.

**Technologies Used:** React.js, Firebase (Firestore & Auth), Tailwind CSS, Cloudinary

#### Expense Tracker | Full-Stack

Demo · GitHub

MERN-stack financial management ecosystem integrating isolated Express controller definitions, clean REST schema architectures, and permanent MongoDB data collection instances.

**Technologies Used:** React, Express.js, MongoDB

Interactive architectural culinary landing module using clean modular styling sheets and targeted dynamic interactive UI rendering scripts.

**Technologies Used:** HTML, CSS, JavaScript, jQuery

## **AWARDS & CERTIFICATIONS**

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- **Frontend Development Program** — CodeGirls Pakistan
- **Internet of Things & Introduction to Python** — Cisco Networking Academy
- **Aspire Leaders Global Fellowship** — Aspire Institute